Li Shen, PhD

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SUMMARY

Ph.D.-trained applied economist focused on demand estimation, market analysis, and economic modeling. Builds multi-million-row market panels and translates empirical evidence and model-based counterfactuals into pricing scenarios, forecasts, and decision-support tools. Combines panel econometrics and causal-inference design with structural modeling, forecast validation, and constrained optimization.

EDUCATION

Clark University — **Worcester, MA**
Ph.D. in Economics — *August 2026*

The State University of New York at Buffalo — **Buffalo, NY**
M.A. in Economics — *May 2019*

Valparaiso University — **Valparaiso, IN**
M.S. in International Economics and Finance — *May 2018*

Chongqing University — **Chongqing, China**
B.Mgmt. in Accounting — *June 2014*

TECHNICAL SKILLS

- **Demand estimation and market analysis:** Price-response and elasticity measurement, high-frequency market panels, market integration, pricing scenarios, constrained optimization, and A/B-test design.
- **Panel econometrics and causal-inference design:** Fixed effects, difference-in-differences, staggered-adoption designs, event studies, shift-share IV, identification and pre-trend diagnostics.
- **Structural and counterfactual modeling:** Quantitative spatial equilibrium, multi-region input-output networks, calibration, structural inversion, labor mobility, and counterfactual simulation.
- **Forecasting and validation:** Seasonal ARIMA, rolling-origin evaluation, forecast-error and spike-risk metrics, sealed-test evaluation, and baseline fallbacks.
- **Programming and data:** Python, SQL, Stata, MATLAB, R, and SAS; reproducible workflows for multi-million-row panel, scanner, mobility, and geospatial data.

SELECTED ECONOMIC RESEARCH

High-Speed Rail and Market Integration — Price Dispersion and Mobility 2021–Present

- Built a 3.99M-observation city-pair-month panel from daily wholesale prices for 10 vegetables across 117 Chinese cities (2014–2022), merged with AutoNavi intercity passenger-mobility indices.
- Used the Green Passage highway-toll exemption and prohibitive HSR freight tariffs for bulk produce to distinguish passenger-mobility and freight-cost explanations for spatial price dispersion.
- Applied pre-trend and identification diagnostics to calibrate the interpretation, retaining descriptive and comparative evidence when a causal claim was not supported.
- Documented 2–5% lower price dispersion among HSR-connected city pairs; mobility controls and cross-product heterogeneity are consistent with passenger mobility and trader interaction rather than freight substitution.

Trade Shock Propagation — Structural Spatial Equilibrium and Quantitative Trade 2023–Present

Job Market Paper; coauthored

- Co-developed a multi-sector, 31-province structural spatial-equilibrium model with inter-provincial input-output networks, non-tradable production, and labor mobility; contributed quantitative model development, calibration-data work, and counterfactual analysis using China's 2017 MRIO, ICIO, and Customs data.
- Derived and implemented alternative labor-mobility formulations, comparing simultaneous location-sector choice with sequential location-then-sector choice through equilibrium allocations and model-implied multipliers.
- Solved province-sector counterfactual equilibria to separate direct export losses from indirect propagation through supplier linkages, non-tradable demand, and worker relocation.
- Counterfactual simulations yield model-implied GDP multipliers from 0.30 to 1.02 across provinces, with local production structure and own-trade share organizing heterogeneity.

APPLIED PROJECTS

Pricing and Promotion Decision Engine 2025–Present

Demand Estimation, Optimization, and A/B Design

Code: github.com/Li-Shen-Clark/pricing-promotion-decision-engine

- Built an end-to-end pricing decision-support application on 6.6M rows of Dominick's Finer Foods scanner data; cleaned to a 4.65M-row panel covering 486 UPCs, 93 stores, and 366 weeks.
- Estimated a log-log fixed-effects price-response model with Duan smearing retransformation to measure historical demand responses to price and promotion changes; the estimates are observational rather than causal.
- Implemented scenario simulations and constrained profit optimization under cost, inventory, and margin constraints; added support guardrails and A/B validation plans with power analysis before rollout.
- Deployed the workflow as a Streamlit app with modular Python code, reproducible requirements, GitHub auto-deployment, and pytest tests.

Vegetable Price Intelligence — Forecasting, Risk, and Model Governance 2026–Present

Portfolio case study: Case Study *Code:* GitHub

- Built an economics-informed price-intelligence system from 8.68M historical wholesale-market records across 117 cities and 30 vegetables, producing 7.36M auditable city-day facts with versioned mappings, quality flags, and coverage tiers.
- Evaluated 7/14/28-day forecasts with rolling-origin validation and sealed final testing; reduced overall WAPE from 18.72% to 17.54% and routed 10 weak product-horizon groups to explicit baseline fallbacks.
- Froze a 14-day price-spike classifier before final evaluation and withheld a proposed directional network when only 4 of 15 frozen edges improved out of sample, making release decisions auditable.

China Population Density Atlas — Interactive Geospatial Visualization 2022–Present

Live app: pop-density-spatial.streamlit.app

- Processed 19 years of 5 km-resolution population-density rasters (2002–2020) covering China into a harmonized annual panel for grid-cell and province-level analysis.
- Built the underlying density, agglomeration, and urban-corridor metrics and deployed an interactive Streamlit application for map-based exploration.

ADDITIONAL ECONOMICS AND PROFESSIONAL EXPERIENCE

Applied Econometric Research and Forecasting 2017–2019

- Built seasonal ARIMA models in SAS to forecast international passenger demand, modeling recurring seasonal patterns for capacity-planning analysis.
- Estimated GDP-demand elasticities and translated alternative economic-growth assumptions into passenger-demand and capacity-planning scenarios.
- Modeled determinants of corporate revenue using regression analysis on time-series and accounting data.

Financial Auditor, Datong Jiaye Property Management Co., Ltd. 2014–2016

- Conducted internal audits, reconciled inventory records, and prepared general-ledger journal entries.

ADDITIONAL QUANTITATIVE EXPERIENCE

Home Credit — Credit Risk Model Stability (Kaggle Silver Medal) Feb–May 2024

- Co-developed a Polars lazy-execution ETL pipeline for multi-depth relational borrower data and engineered max/last/mean aggregation features to capture historical behavior across linked tables.
- Optimized LightGBM/CatBoost ensembles for the Gini Stability metric, which penalized predictive deterioration over time; used correlation-based feature grouping and type downcasting to reduce memory use by more than 50%.

ICR — Identifying Age-Related Conditions (Kaggle Bronze Medal) May–Aug 2023

- Designed subgroup-aware resampling with auxiliary "Greek" metadata, applying RandomOverSampler within Alpha-defined latent groups to preserve population structure under severe class imbalance.
- Co-developed an XGBoost/TabPFN voting ensemble evaluated under a custom Balanced Log Loss metric and selected validation-based decision thresholds to manage false negatives.

TEACHING AND HONORS

- **Teaching Assistant, Clark University** (2020–2025): Led graduate finance and computational sessions covering Monte Carlo methods, stochastic processes, and derivatives; supported undergraduate macroeconomics courses.
- **Honors:** Sheftel Award for Research Excellence (two-time recipient, 2025); Veendorp Best Field Paper Award (2022); Doctoral Fellowship (2019–2026).